



ATTACK DETECTION IN CLOUD COMPUTING USING MACHINE LEARNING

```
main (
    static String AppName = "SQL Mail";
    static String AppVersion = " 0.0.1 ";
    static String AppAuthor = "Jeffrey Cobb";
    static String AppDate = "August 8th, 2007";
    static String AppPath = System.getProperty("user.dir");
    static String AppDriver = "smallsql.database.SSDriver";
    static String AppDBHeader = "jdbc:smallsql:";
    static String AppDBPath = AppPath + "/sqlmail";
    static String AppPreferences = AppPath + "/sqlmail_prefs";
    /** Creates a new instance of Main */
    public Main() {
    }

    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) throws Exception {
        // TODO code application logic here

        boolean bDBConnect = false;
        int result = 0;
        frmSQLMailForm = new frmMain();
        frmSQLMailForm.println("\r\n" + AppName + "\r\nVersion" + AppVersion + "\r\nAuthor: "
            + AppAuthor + "\r\n");

        frmSQLMailForm.getDefaultToolkit();
        frmSQLMailForm.setSize(
            frmSQLMailForm.getWidth() + " --- " + screen.get
```



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INTRODUCTION

CloudAtk-ML is a smart system that detects cyber-attacks in cloud networks using Machine Learning. It uses XGBoost for known attacks and Isolation Forest for new attacks. The system analyzes network traffic in real time, explains its decisions using SHAP, and stores results in a MySQL database through a Flask web interface.



PROBLEM STATEMENT

Modern cloud environments face increasing and sophisticated security threats. Traditional signature-based security systems fail to detect new and advanced attacks (Zero-day attacks). The complexity and dynamism of cloud infrastructure make manual traffic monitoring ineffective.



MAIN OBJECTIVE

Develop a smart and complete system to detect cyber-attacks in cloud environments using advanced Machine Learning techniques, achieving high detection accuracy, identifying both known and new attacks in real time, and providing full transparency in security decisions.

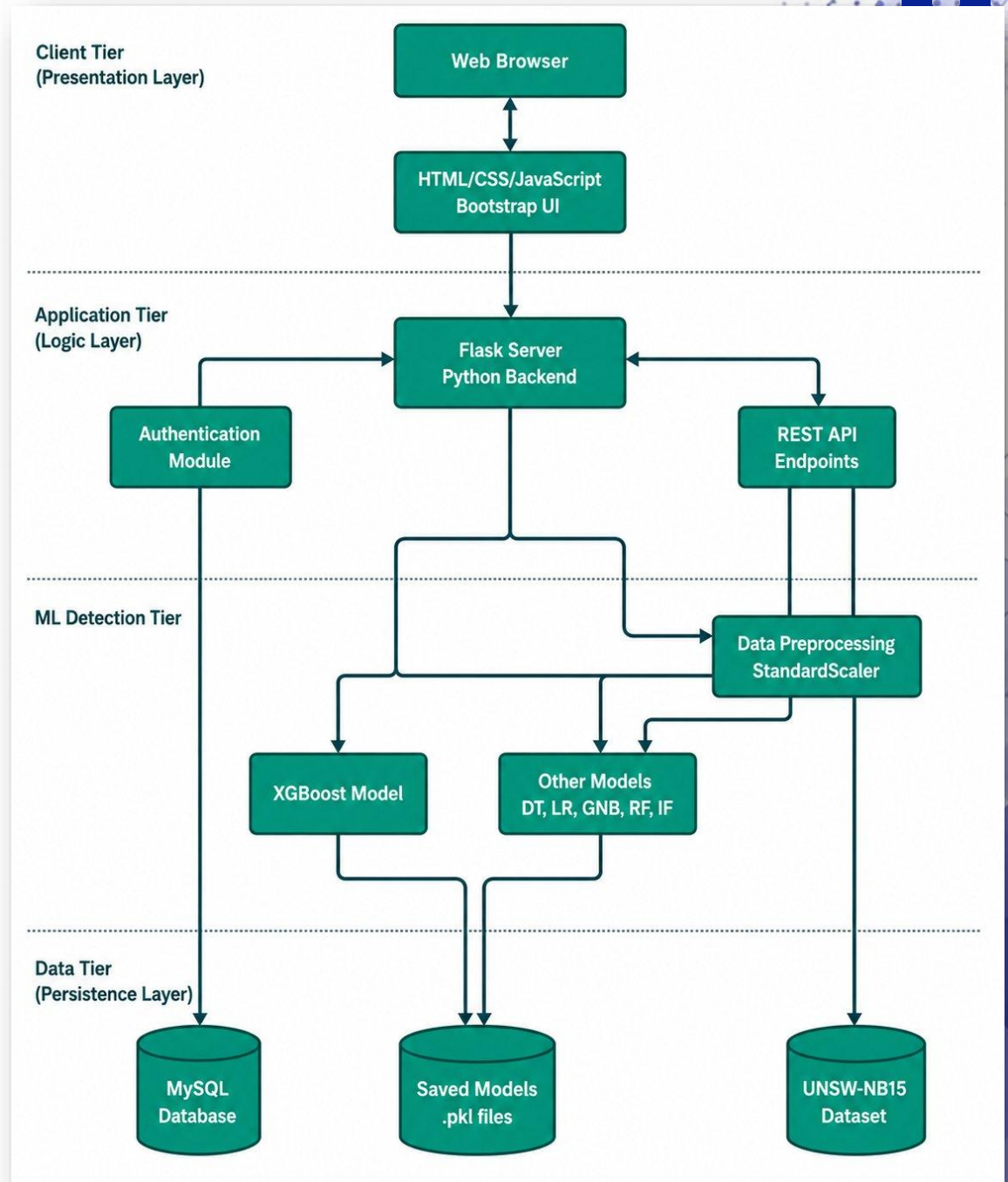


PROPOSED SOLUTION

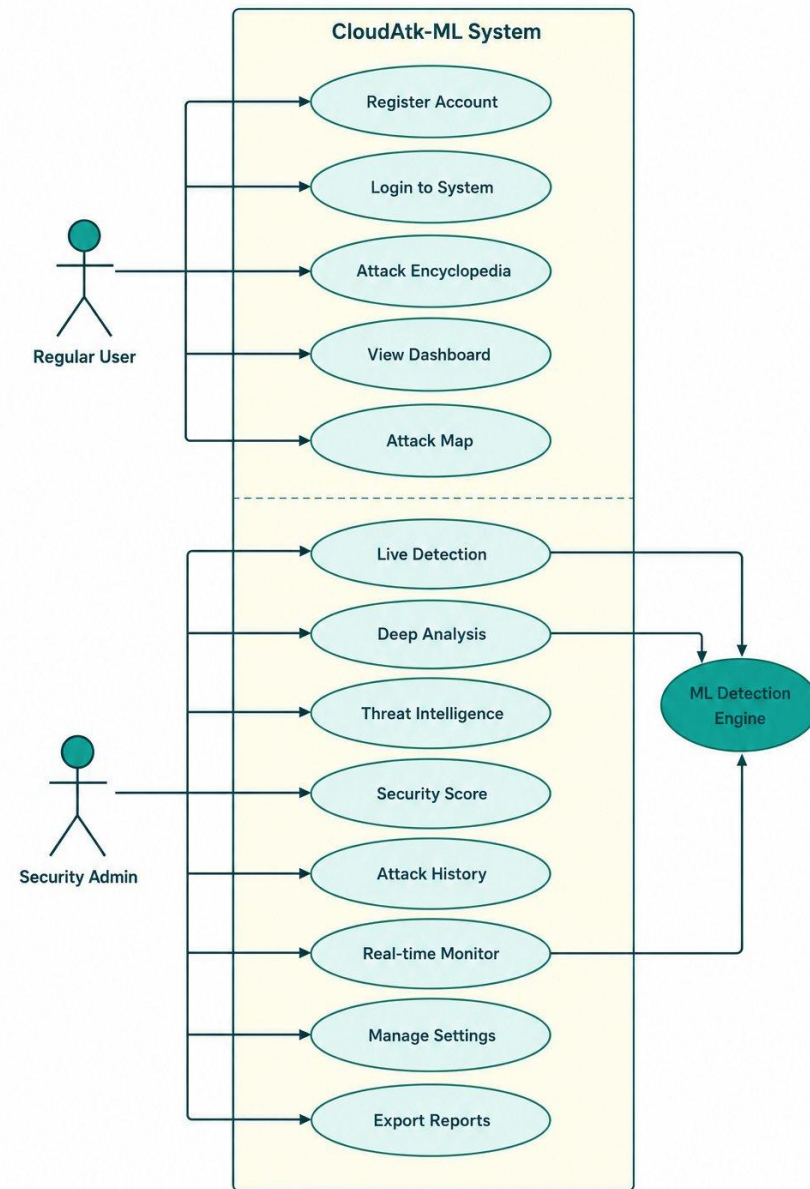
The proposed solution is to build an intelligent cloud security system that uses Machine Learning to detect both known and new cyber-attacks in real time. The system combines XGBoost for accurate attack classification and Isolation Forest for anomaly detection.



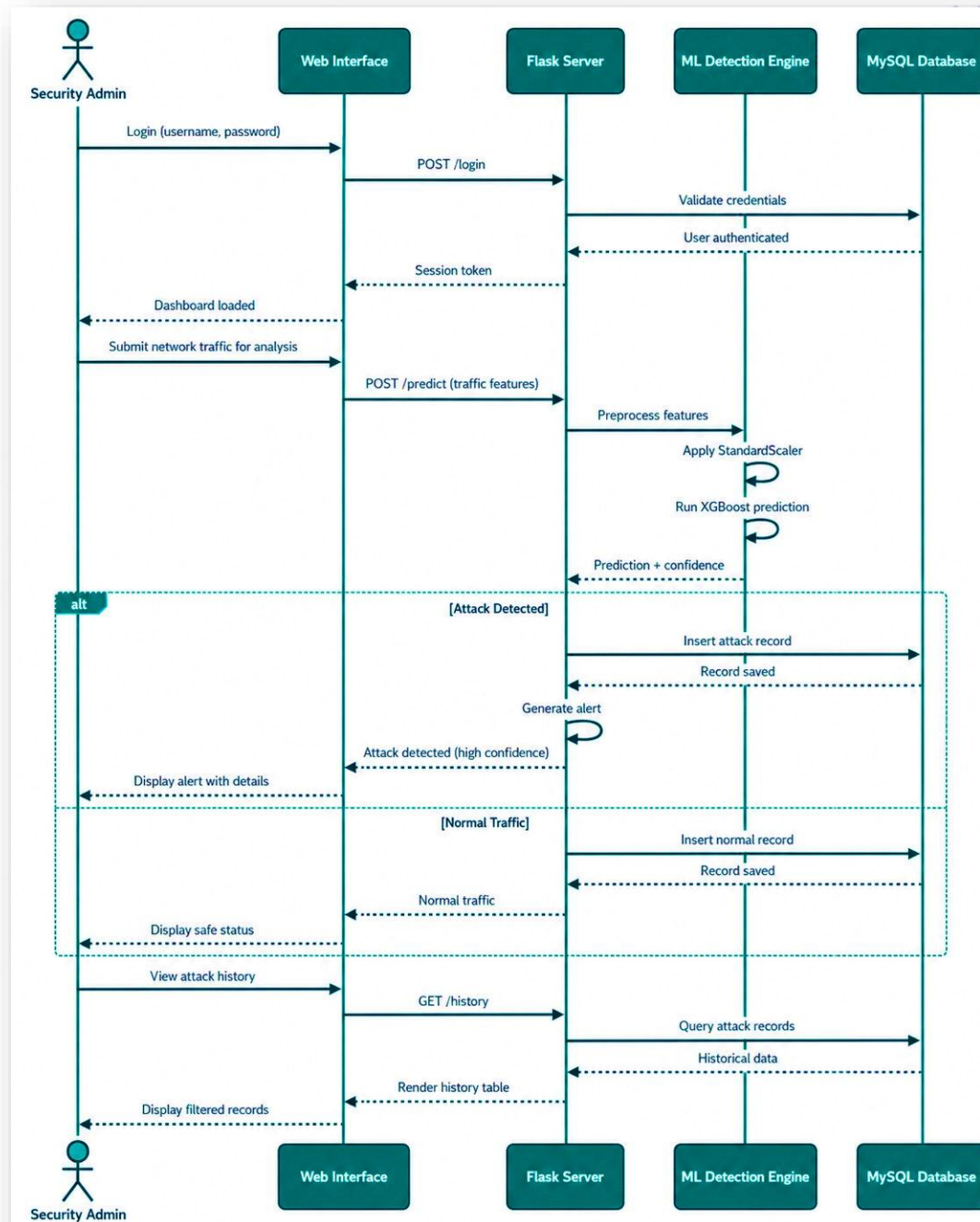
SYSTEM ARCHITECTURE



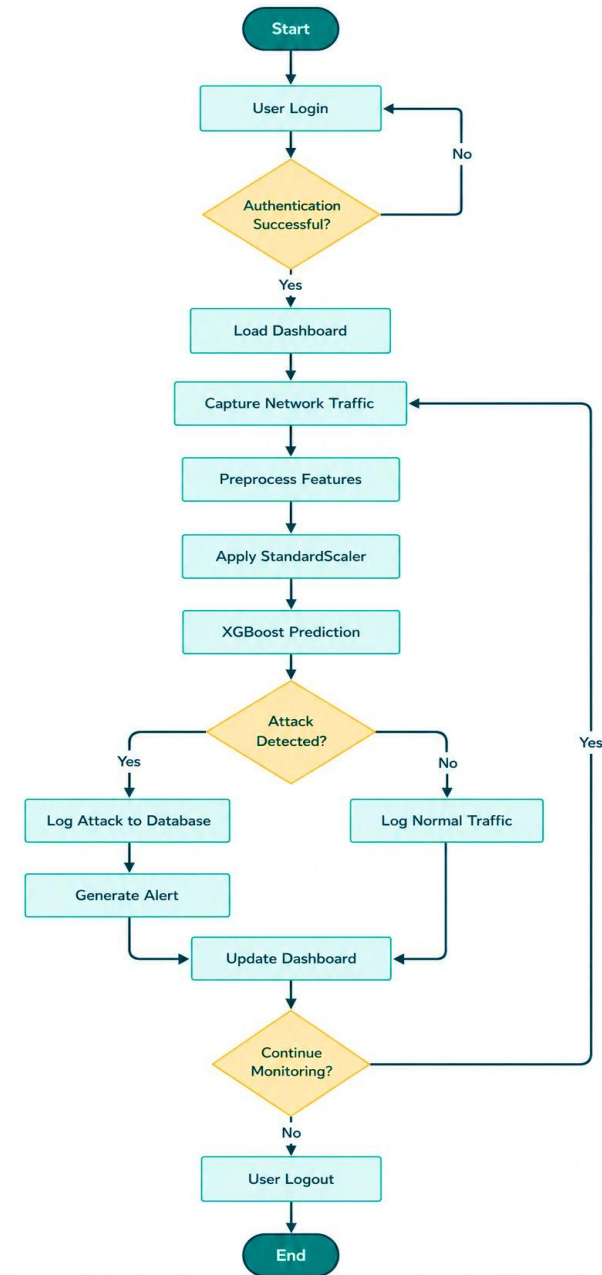
USE-CASE DIAGRAM OF CLOUDATK-ML



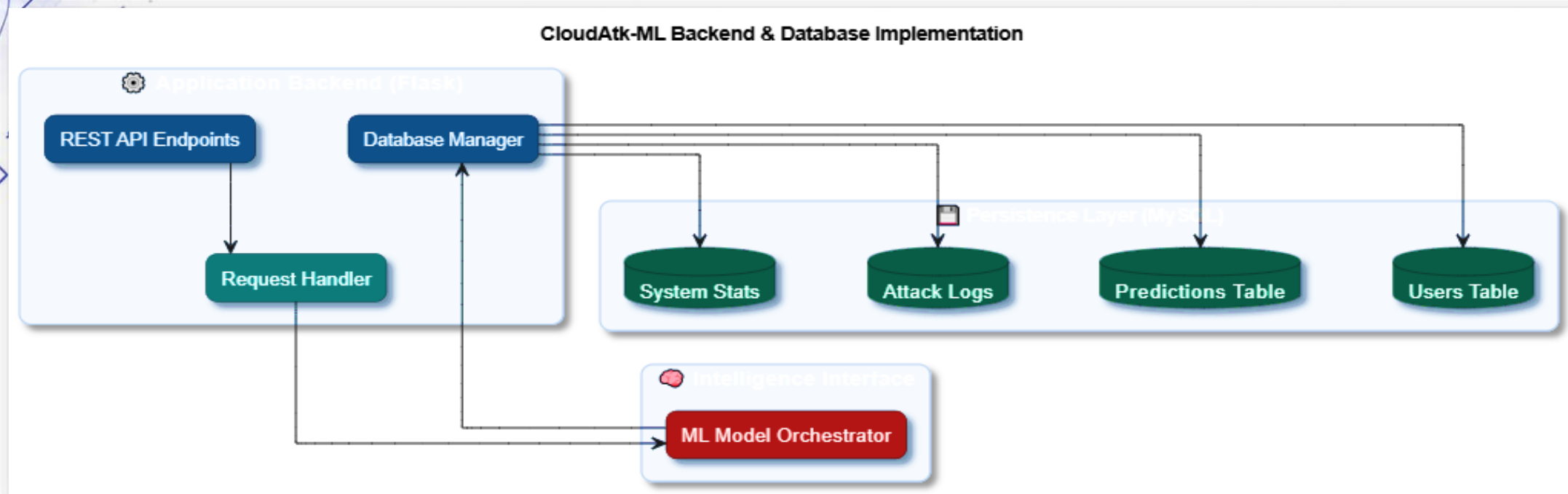
SEQUENCE DIAGRAM FOR ATTACK DETECTION FLOW



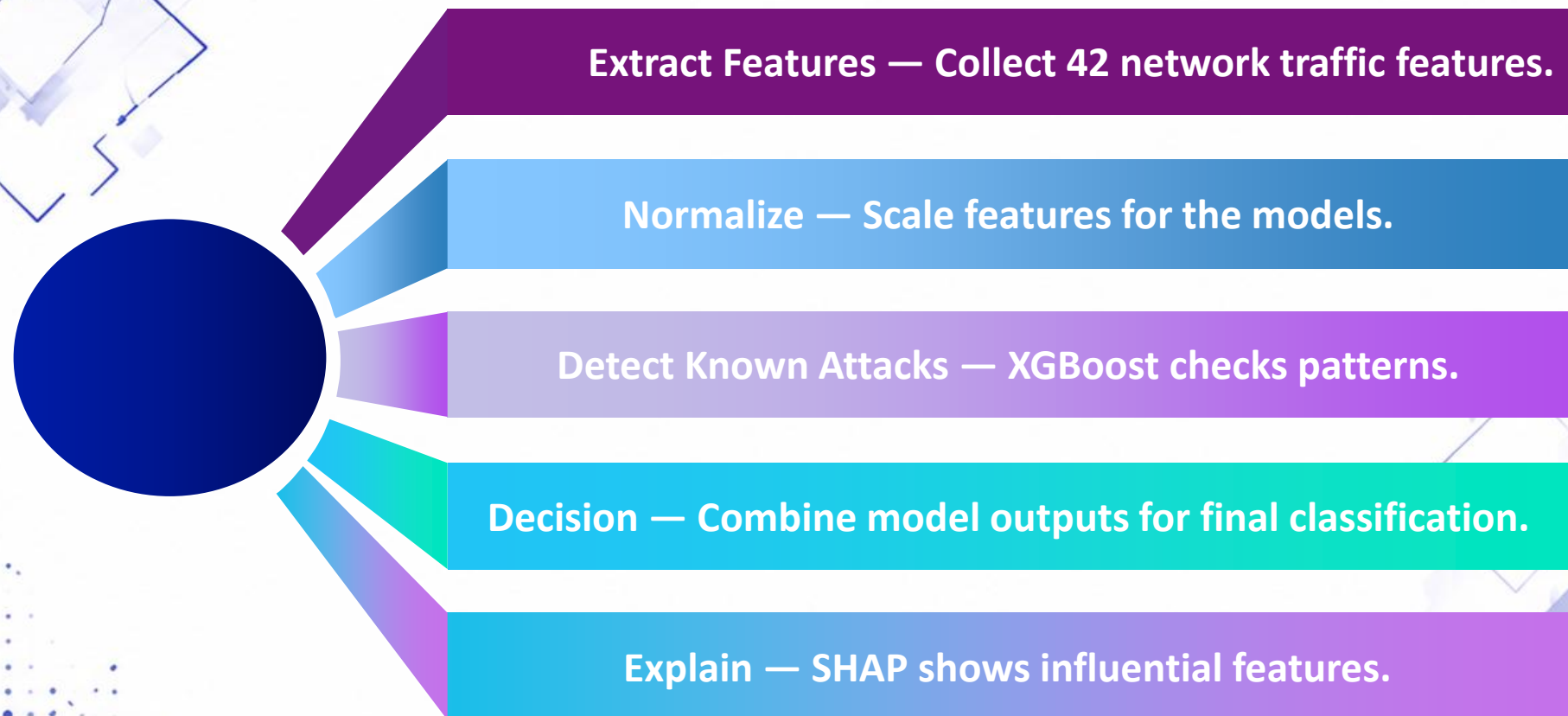
ACTIVITY DIAGRAM OF CLOUDATK-ML



BACKEND AND DATABASE IMPLEMENTATION



CORE FUNCTIONALITY OF CLOUDATK-ML



SOFTWARE REQUIREMENTS

HTML

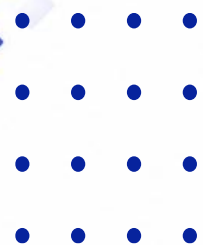


CSS

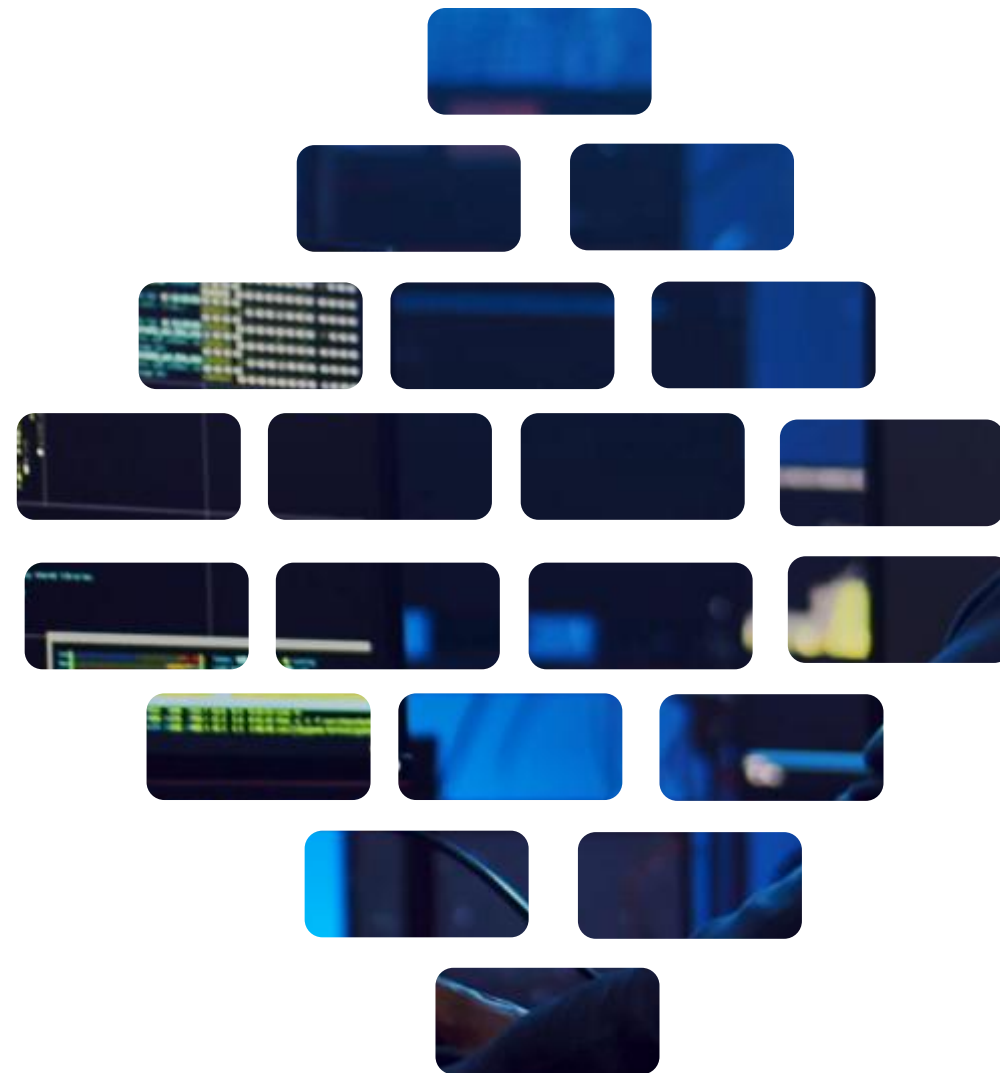


JavaScript

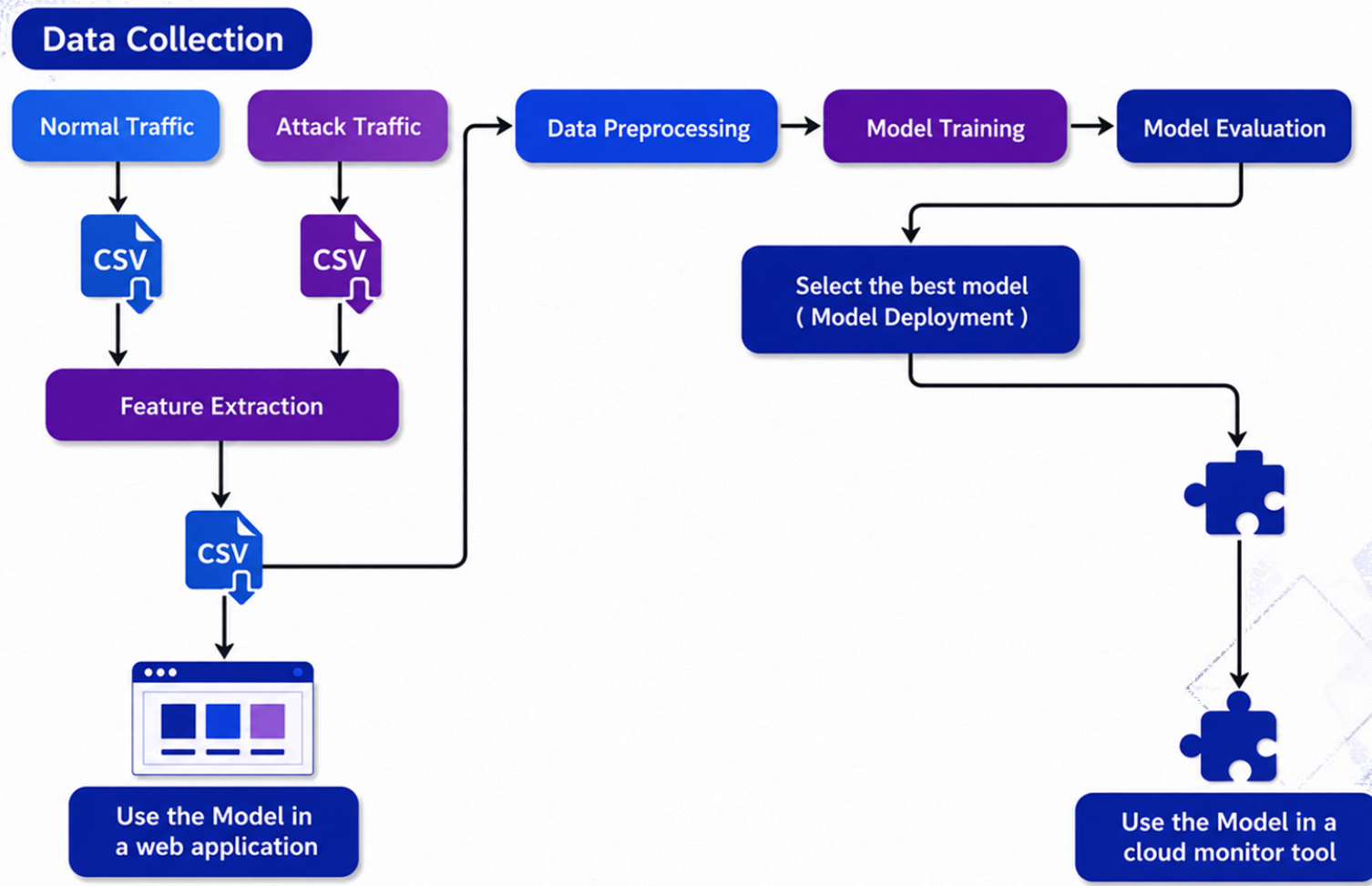




METHODOLOGY



DATA METHODOLOGY & PROCESSING PIPELINE



MACHINE LEARNING ALGORITHMS

1

XGBoost (Primary Model)

2

Isolation Forest

3

SHAP (Explainable AI)

4

Hybrid System

ATTACK TYPES COVERED BY THE SYSTEM

Based on the UNSW-NB15 dataset:

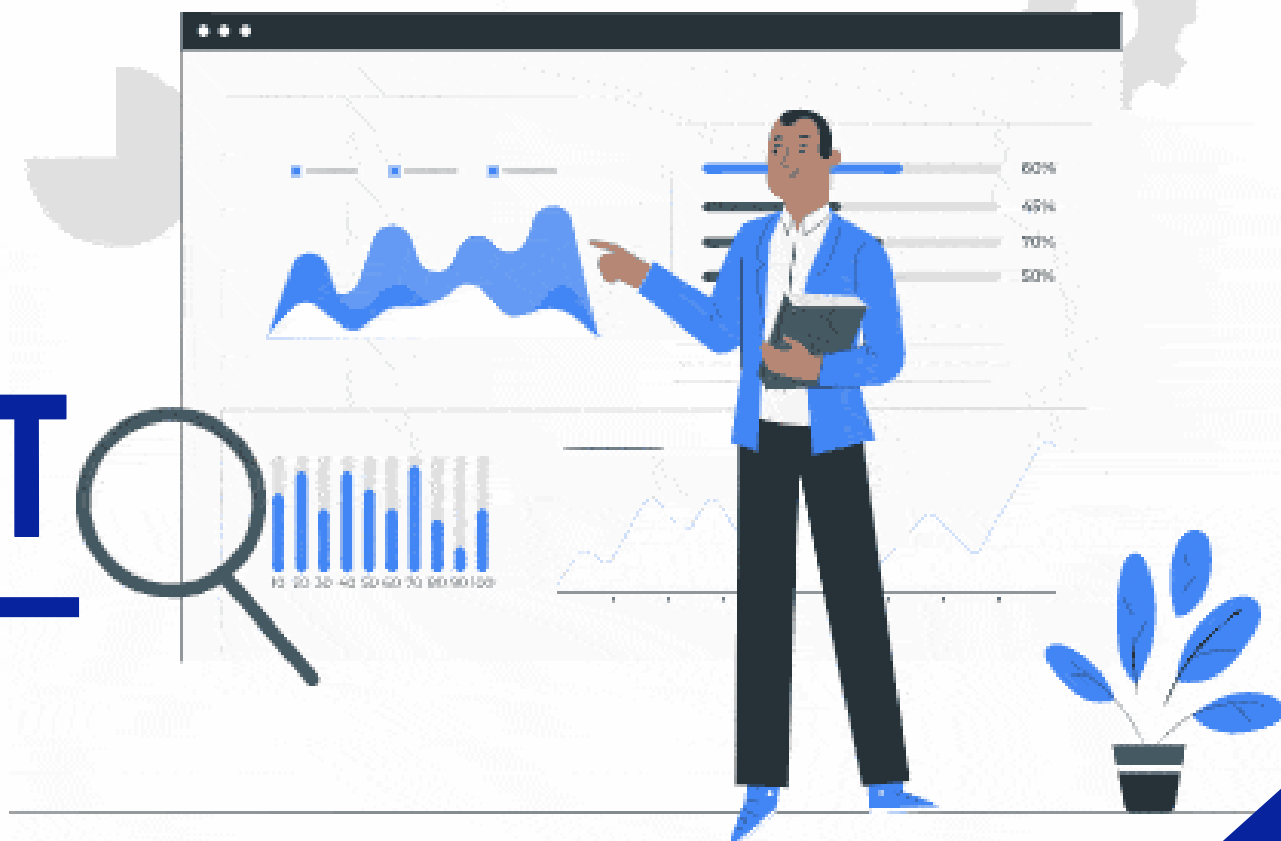
- **Generic** — Common attacks that don't fit a specific category, targeting general network vulnerabilities.
- **Exploits** — Attacks that leverage known software vulnerabilities to gain unauthorized access or execute malicious code.
- **Fuzzers** — Sending random or malformed data to crash systems and discover hidden vulnerabilities.
- **DoS (Denial of Service)** — Overwhelming network resources to make services unavailable to legitimate users.
- **Reconnaissance** — Scanning and probing the network to gather information before launching an attack.

ATTACK TYPES COVERED BY THE SYSTEM

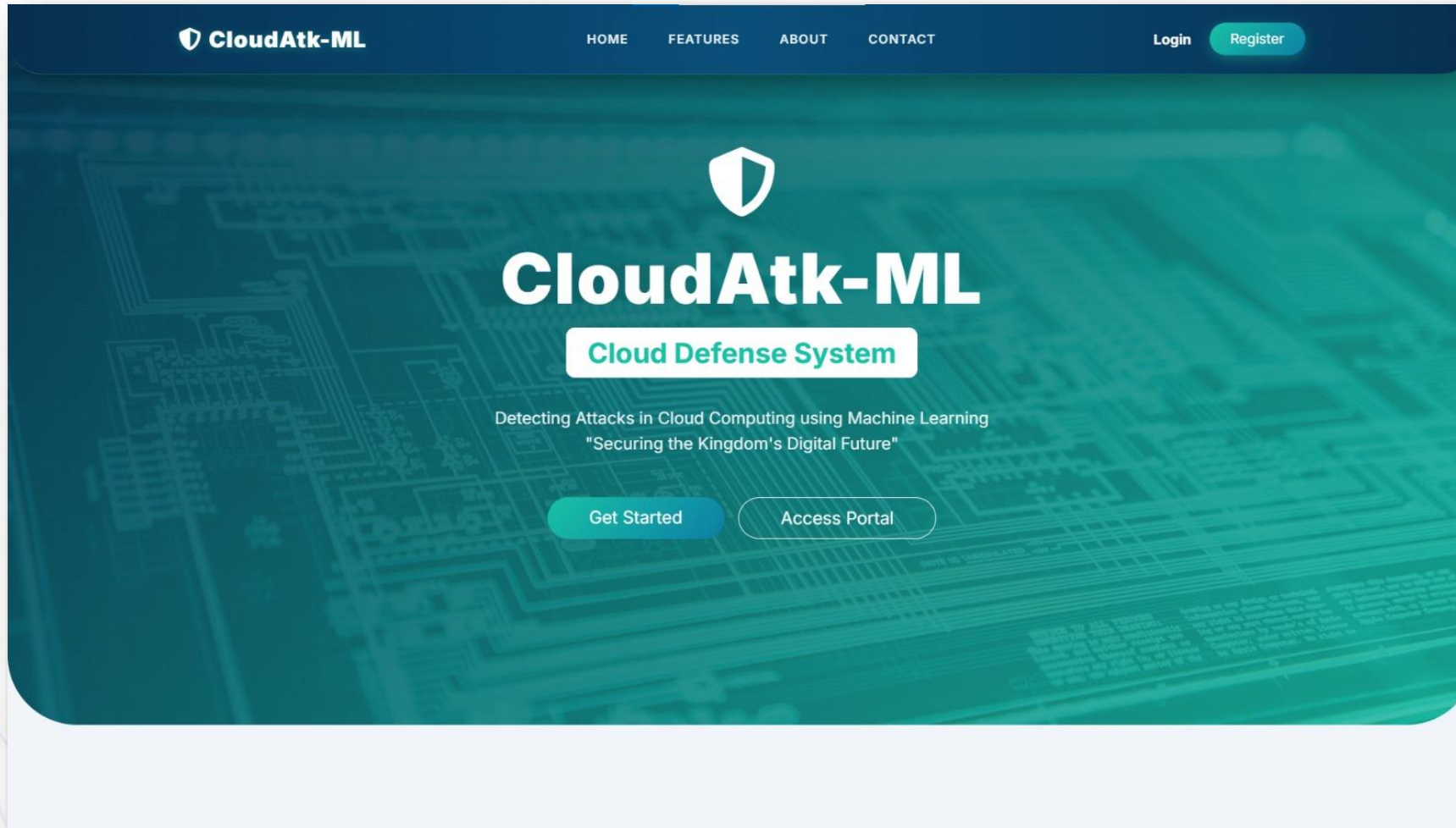
Based on the UNSW-NB15 dataset:

- **Analysis** — Malicious deep inspection of network traffic to extract sensitive data or identify weaknesses.
- **Backdoor** — Hidden access mechanisms planted in systems to allow unauthorized remote control.
- **Shellcode** — Injecting and executing low-level machine code to gain control over a vulnerable system.
- **Worms** — Self-replicating malware that spreads automatically across networks without user interaction.

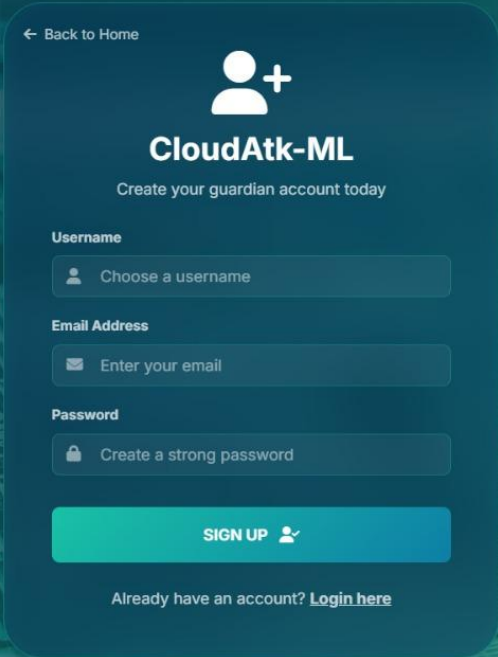
THE RESULT



HOME PAGE



REGISTRATION



← Back to Home



CloudAtk-ML

Create your guardian account today

Username

 Choose a username

Email Address

 Enter your email

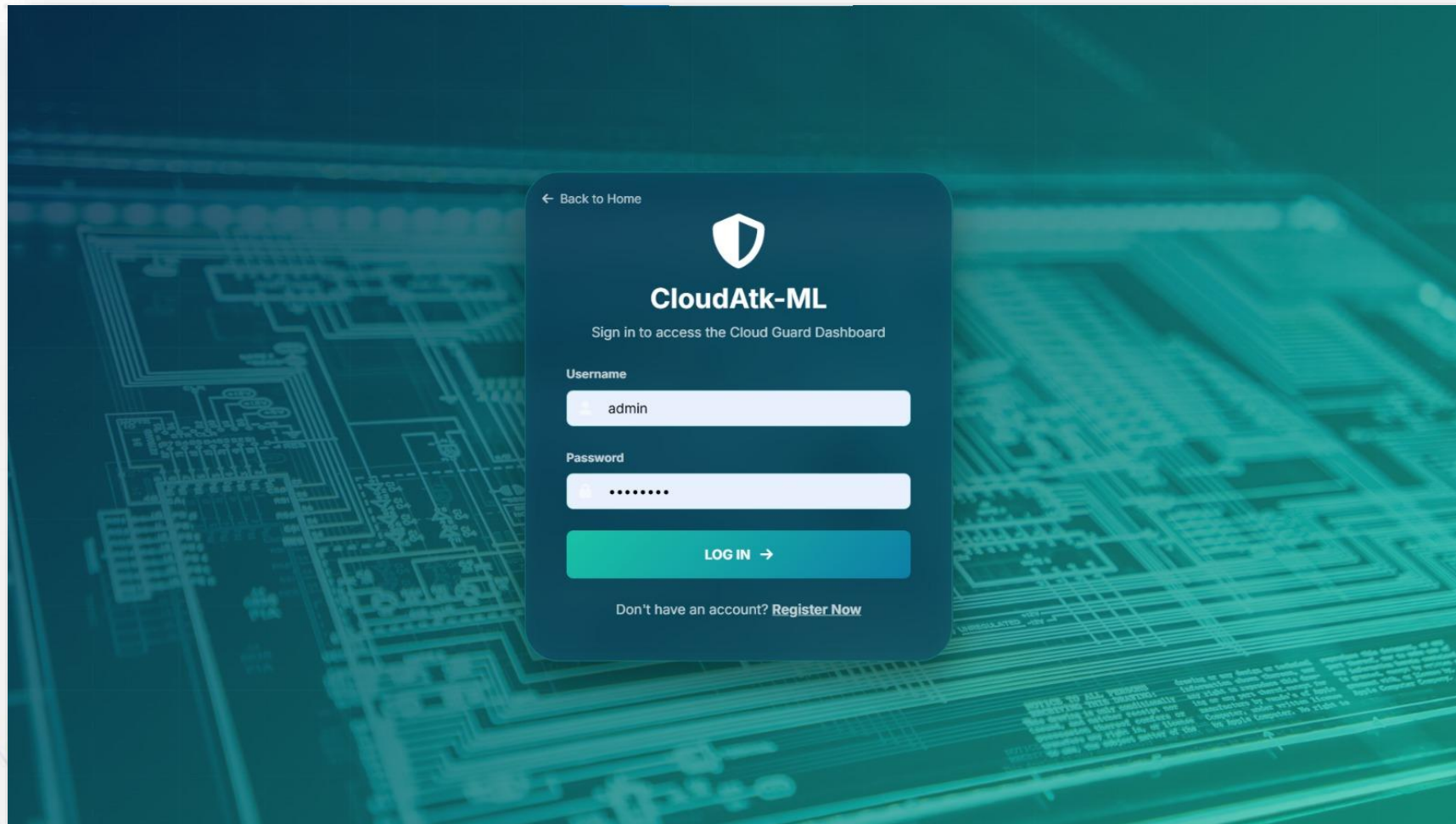
Password

 Create a strong password

SIGN UP 

Already have an account? [Login here](#)

LOGIN

The image shows a login interface for CloudAtk-ML. It features a dark teal background with a faint circuit board pattern. A central white rounded rectangle contains the login form. At the top left of this rectangle is a back arrow and the text 'Back to Home'. Below this is a shield icon and the text 'CloudAtk-ML'. Underneath is the instruction 'Sign in to access the Cloud Guard Dashboard'. The form includes a 'Username' field with a user icon and the text 'admin', and a 'Password' field with a lock icon and masked dots. A green 'LOG IN →' button is positioned below the password field. At the bottom of the form is the text 'Don't have an account? Register Now' with 'Register Now' as a link. The entire interface is framed by decorative blue and white geometric patterns on the left and right sides.

← Back to Home



CloudAtk-ML

Sign in to access the Cloud Guard Dashboard

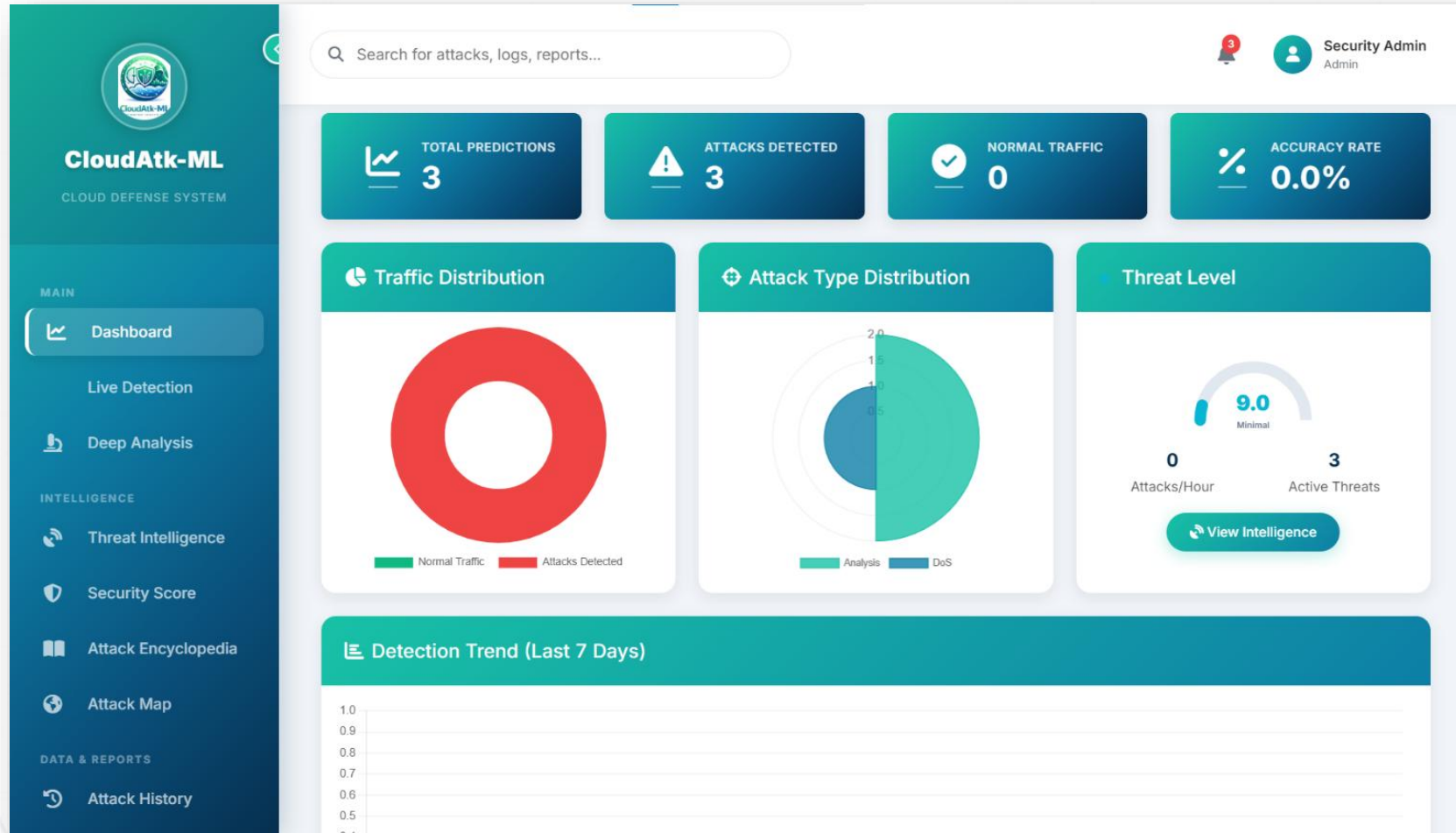
Username

Password

LOG IN →

Don't have an account? [Register Now](#)

MAIN DASHBOARD



LIVE ATTACK DETECTION



CloudAtk-ML
CLOUD DEFENSE SYSTEM

MAIN

- Dashboard
- Live Detection**
- Deep Analysis

DATA & REPORTS

- Attack History
- Real-time Monitor
- XAI Reports

SYSTEM

- Settings
- Logout

Q

Search for attacks, logs, reports...

3





Security Admin
Administrator

Live Attack Detection

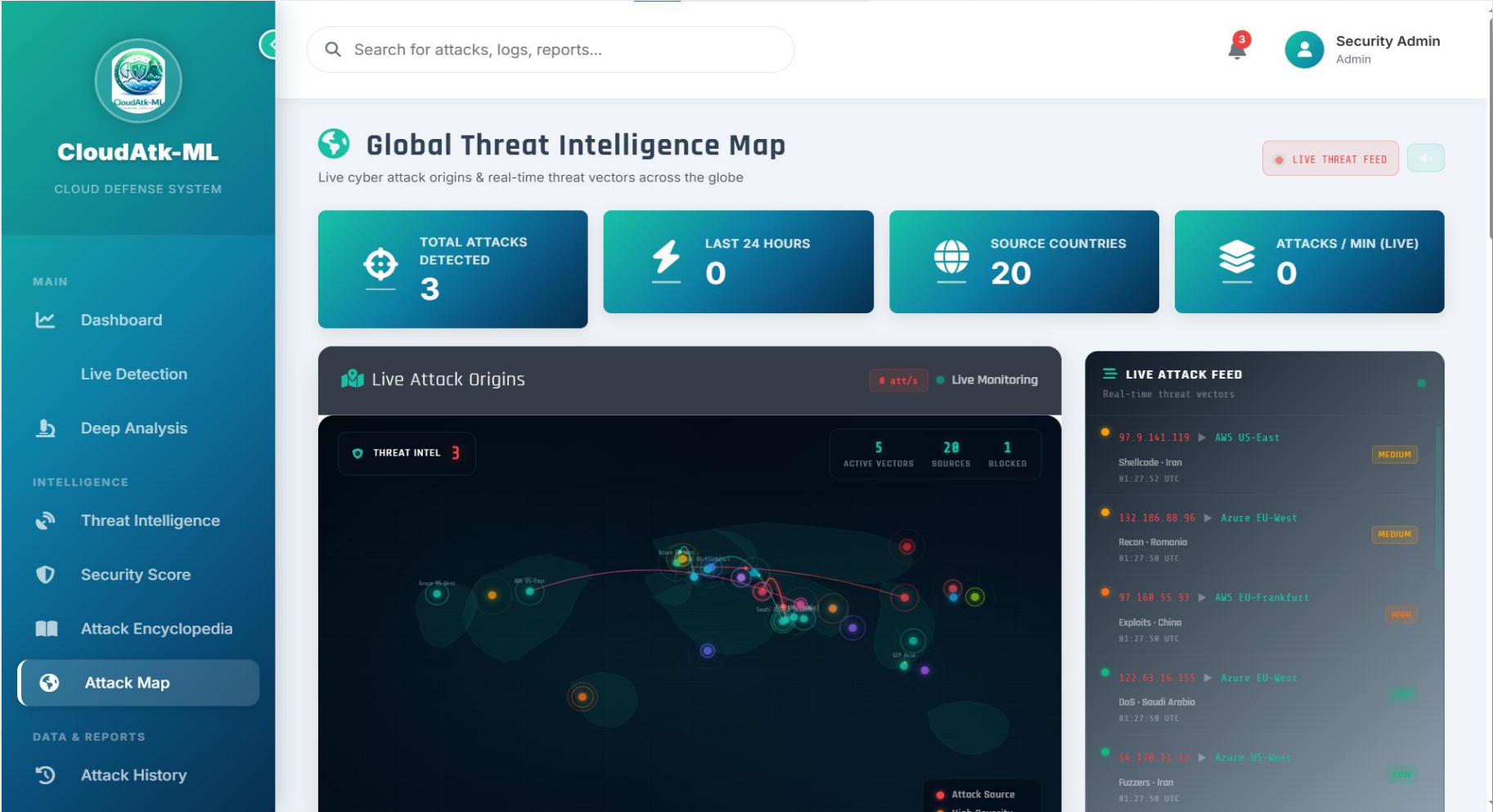
Enter network traffic features to detect potential security threats in real-time

 Network Traffic Features

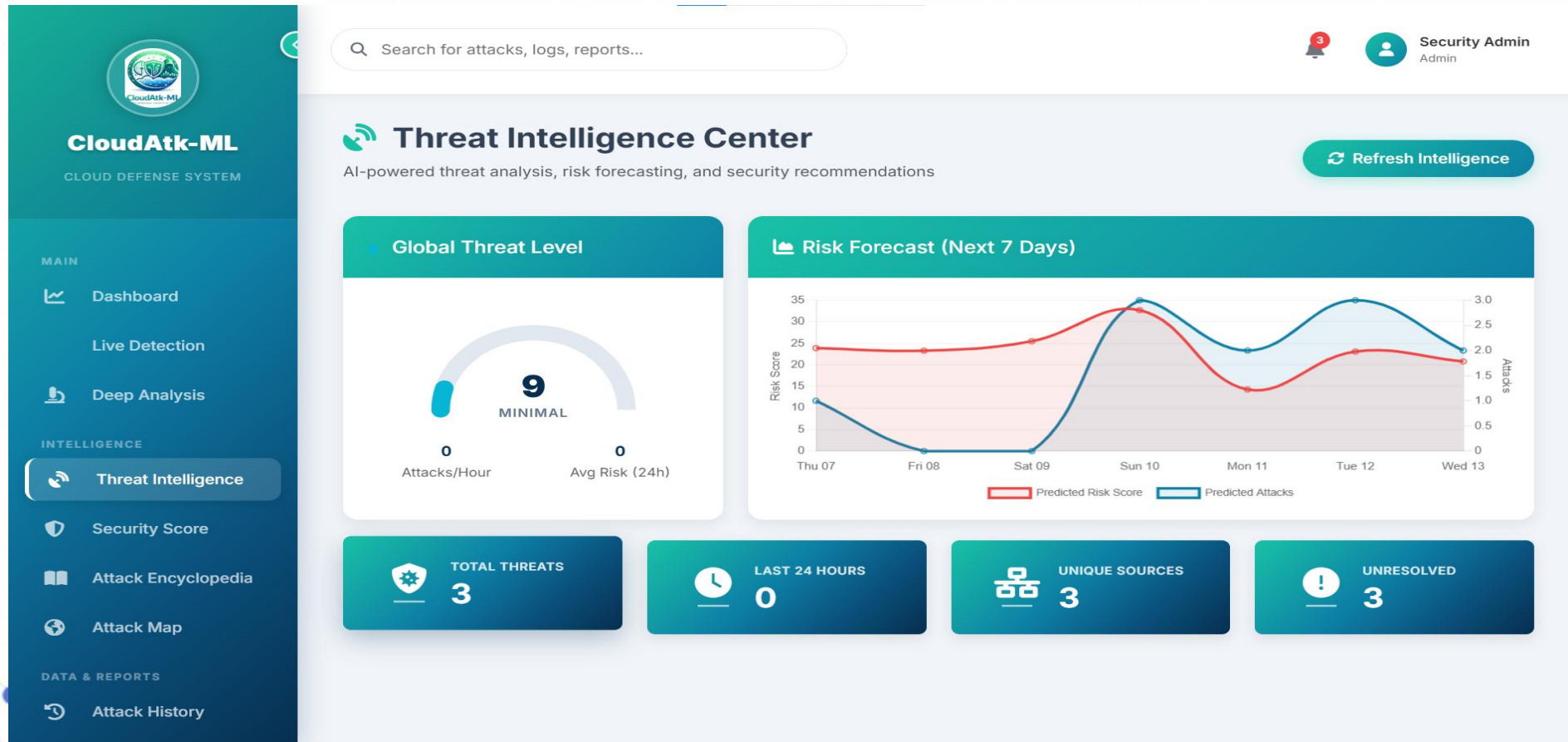
Load Sample Data

dur	proto	service	state
spkts	dpkts	sbytes	dbytes
rate	sttl	dttl	sload
dload	sloss	dloss	sinpkt
dinpkt	sjit	djit	swin
stcpb	dtcpb	dwin	tcprtt
synack	ackdat	smean	dmean
time_delta	max_size_bytes	rtt_mean	rtt_stddev

GLOBAL THREAT INTELLIGENCE MAP



THREAT INTELLIGENCE CENTER



REAL-TIME NETWORK MONITOR



CloudAtk-ML
CLOUD DEFENSE SYSTEM

MAIN

- Dashboard
- Live Detection
- Deep Analysis

INTELLIGENCE

- Threat Intelligence
- Security Score
- Attack Encyclopedia
- Attack Map

DATA & REPORTS

- Attack History
- Real-time Monitor

Search for attacks, logs, reports...



Security Admin
Admin

Real-time Network Monitor

Live monitoring of network traffic, security threats & system telemetry

IDS: **ACTIVE** Firewall: **RUNNING** Threat Level: **ELEVATED** ML Models: **LOADED** CPU: 48 % MEM: 52 % PKT/s: 1.3k

UTC: 01:20:33



SYSTEM STATUS
Active



TOTAL ANALYZED
3



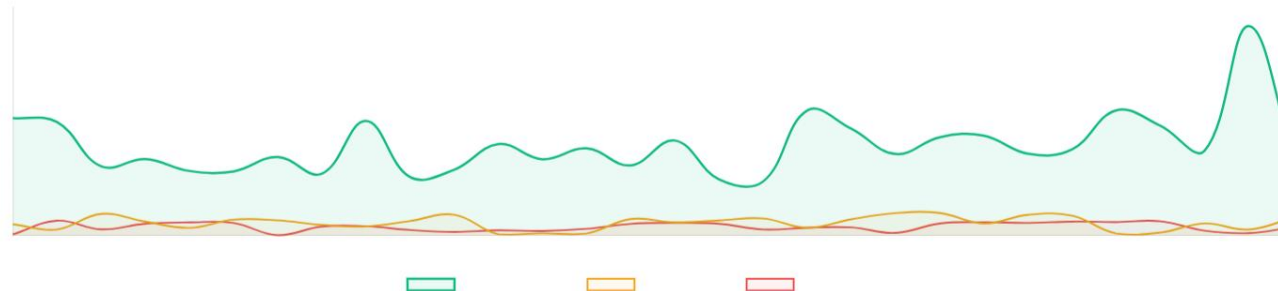
ACTIVE THREATS
3



NETWORK TRAFFIC
64.7 Mbps

Live Traffic Analysis

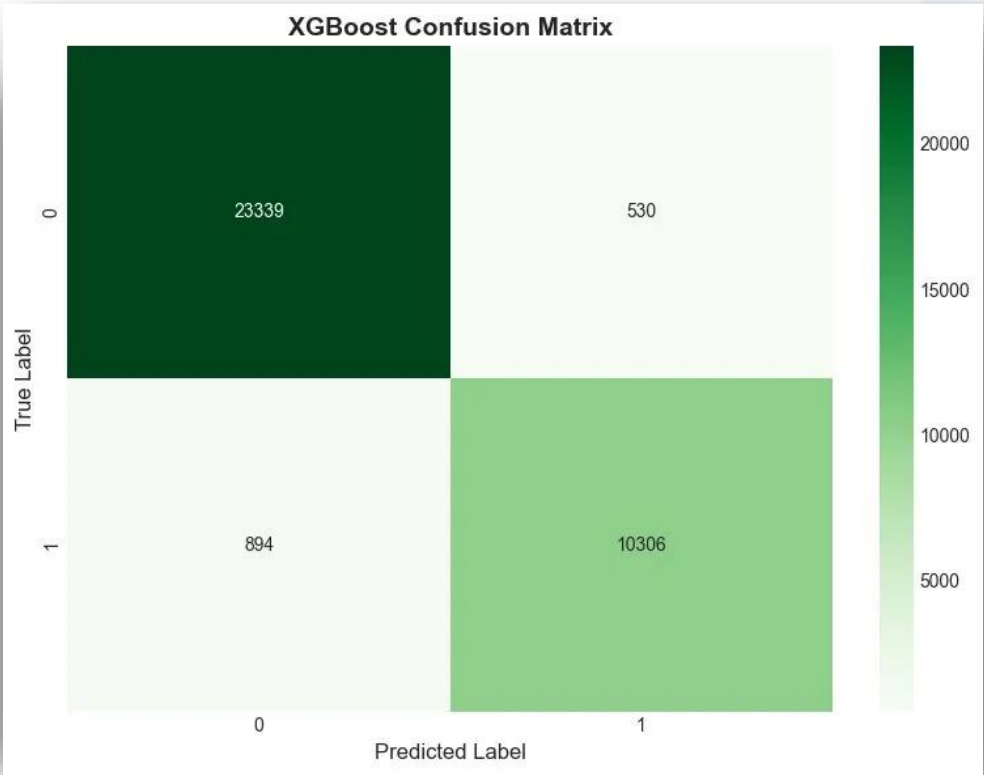
64.7 Mbps **Live**



XGBOOST PERFORMANCE METRICS

XGBoost Classification Report:

	precision	recall	f1-score	support
Normal	0.96	0.98	0.97	23869
Attack	0.95	0.92	0.94	11200
accuracy			0.96	35069
macro avg	0.96	0.95	0.95	35069
weighted avg	0.96	0.96	0.96	35069



CONCLUSION

CloudAtk-ML successfully proved that combining Machine Learning with a full web application can effectively detect cyber attacks in cloud environments. Six models were trained and tested on the UNSW-NB15 dataset. Random Forest achieved the highest accuracy at 96.03%, followed by XGBoost at 95.94%, which was chosen as the main model due to its speed and explainability via SHAP. The final result is a complete, ready-to-use security dashboard for SOC analysts.

RECOMMENDATIONS & FUTURE WORK

❑ RECOMMENDATIONS

- Use newer datasets alongside UNSW-NB15 to improve model accuracy
- Add two-factor authentication to make login more secure
- Use a production server like Gunicorn instead of Flask's built-in server
- Monitor the model regularly to catch any drop in performance over time
- Encrypt the database to protect stored user and attack data

❑ FUTURE WORK

- Real-time traffic monitoring using Zeek/Suricata.
- Add deep-learning models (CNN, LSTM, Transformer).
- Enhance SHAP explainability features.
- Implement drift detection and automatic retraining.
- Integrate with SIEM systems (Splunk, QRadar, Sentinel).
- Develop a mobile app with alerts and monitoring.

The slide features decorative geometric patterns in the corners. The top-left corner has overlapping squares and diamonds in various shades of blue. The top-right corner has a grid of small blue dots and a line with a dot. The bottom-left corner has concentric circular arcs and a grid of small blue dots. The word "THANKS" is centered in a large, bold, dark blue font.

THANKS

The slide features decorative geometric patterns in the corners. The top-left corner has overlapping squares and diamonds in various shades of blue. The top-right corner has a grid of small blue dots and a line with a dot. The bottom-left corner has concentric circular arcs and a grid of small blue dots. The bottom-right corner is mostly empty, with a small blue dot near the center.

Any Questions ?